Problem 1

$$F(x,y,z) = x^2 + xy + 2y^2 + z^2, \qquad P = (1,1,2), \quad Q = (3,0,4).$$

1. Find $\nabla F(P)$ and the directional derivative of F at P in the direction from P to Q.
2. Find the unit direction of fastest increase of F at P, and the maximum rate of change.
3. Find the tangent plane and a vector equation of the normal line at P to the level surface of F through P.

Problem 2

$$f(x,y) = x^3 + y^3 - 6xy, \qquad R = [0,3] \times [0,3].$$

1. Find all critical points of f on $\mathbb{R}^2$, classify them using the Second Derivatives Test, and give their function values.
2. Find the minimum and maximum values of f on each of the four boundary edges of R, including the endpoints.
3. Find the absolute minimum and maximum of f on R, and all points where they occur.

Problem 3

$$f(x,y) = xy, \qquad g(x,y) = x + y = 2.$$

1. Use Lagrange multipliers to find all candidate points.
2. Calculate $f(3,-1)$ and $f(4,-2)$. Use $f(x, 2-x) = x(2-x)$ to decide whether a minimum value exists.
3. Substitute $y = 2 - x$ and write $f(x, 2-x) = 1 - (x-1)^2$. Find the maximum value and where it occurs.